IMU-Based Gait Analysis In Horses Using Equi-Pro

Project Description

- This project analyzes walk, trot, and canter gaits in horses of various breeds using EquiMoves wireless IMUs.
- Gait asymmetries from past injuries may indicate ongoing pain or discomfort.
- A Polar heart rate monitor was briefly used to explore links between heart rate and moments of pain during movement.
- The goal is to assess how gait type relates to asymmetry and physiological response.
- Data from all sessions must be carefully synchronized by time, stride, or other methods, for meaningful analysis.

Introduction

- Lameness is a common and costly issue in horses, typically assessed through subjective visual exams.
- EquiPro uses IMUs for objective lameness detection. [1]
  - IMUs capture detailed movement data.
  - EquiMoves software provides in-depth gait analysis.
  - Access to processed accelerometer data will enable mathematical analysis in our research.
- Polar V800 heart rate monitor is a non-invasive tool to collect inter-beat interval data.
- Hexoskin can be used for measuring human heart rate.

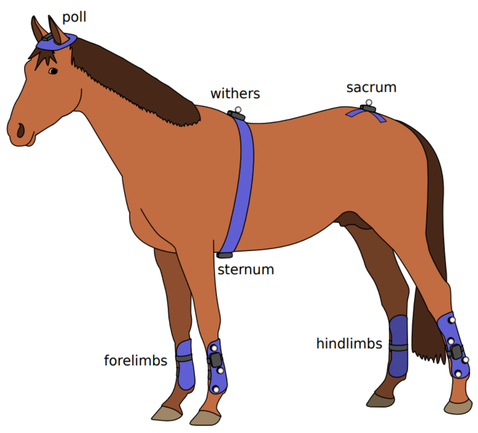

Figure 1: Placement of EquiPro IMUs on horse. [1]

Methods

- Horses were lunged at various gaits and directions in the indoor arena.
- Seven wireless, time-synced IMUs (precision: 100 ns) were placed on the key body points to capture 3D motion.
- A Polar V800 heart rate monitor (chest strap + receiver) was used in select sessions to collect heart rate data.
- All data streams must be synchronized via timestamp or stride.

Results

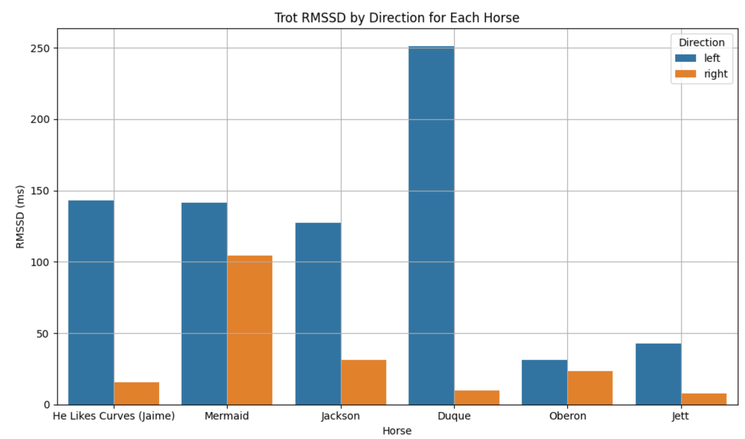

Figure 2: Root Mean Square of Successive Differences between RR intervals for each horse trotting left and right during one session.

Horse	Recovery (%)
Oberon	68.80%
Jaime	59.17%
Jett	49.72%
Jackson	29.19%
Mermaid	17.45%
Duque	11.41%

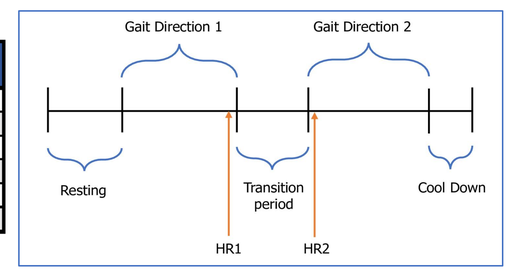

Figure 3: Table with relative recovery of each horse and diagram of lunging session with metrics used to calculate recovery.

Future Work

- Our immediate priority is to establish a post-session data-export workflow for each lunging session.
- Because EquiMoves, Polar, and Hexoskin are independent systems, we need to sync their streams in near real-time to create a time-aligned data matrix with one row corresponding to a single moment in time.
- Our long-term goal is to build a comprehensive model of mutual horse-human interactions.

Glossary of Technical Terms

Lameness: an alteration of the normal gait due to a functional or structural disorder, commonly attributed to orthopedic pain. [1]
Walk: a slow, symmetrical, four-beat gait in which each hoof contacts the ground at separate, evenly spaced intervals.
Trot: a faster, symmetrical, two-beat gait where opposite diagonal hooves strike the ground simultaneously in an alternating pattern.
Canter: a fast, asymmetrical, three-beat gait characterized by one hind hoof hitting the ground first, followed by the opposite diagonal pair, and then the leading front hoof.

References

1. Harder, V., Serra Bragança, F. M., Rhodin, M., Voskamp, J. P., Van Weeren, P. R., Roepstorff, L., & Barneveld, A. (2019). EquiMoves: A wireless networked inertial measurement system for objective examination of horse gait. Sensors, 19(2), 480. https://doi.org/10.3390/s19020480

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